

Stony Brook University
Instructor: Alejandro Melo Ponce

Spring 2019

Midterm Exam I

Instructions: Answer all three questions. The total number of points for the exam is 100. Each question has an equal weight. Answer the questions only in the **blue books provided and be sure to write your name and ID number in each blue book you use**. This exam is closed notes/books. Calculators are allowed but other electronic devices are **not**. For questions with a numerical answer, give a numerical answer. Good luck!

1. Consider the utility function

$$u(x_1, x_2) = x_1^{\frac{1}{2}}(x_2 + 1)^{\frac{1}{2}} \quad x_1 \geq 0, x_2 \geq 0 \quad (1)$$

with the accompanying budget constraint:

$$p_1 x_1 + p_2 x_2 \leq M, \quad p_1 > 0, p_2 > 0, M > 0. \quad (2)$$

- (a) Fix a given utility level $\bar{U} > 0$ and find an explicit expression for the indifference curve defined by the utility level $\bar{U} > 0$. Then, derive an explicit expression for the marginal rate of substitution between good x_1 and good x_2 .
 - (b) Draw the indifference curve (for this associated level of utility $\bar{U}$) and carefully label the graph and its elements.
 - (c) Show that the utility function is strictly increasing (and hence monotone) in x_1 and x_2 .
 - (d) State (*but do not solve*) the utility maximization problem and briefly describe its content, in particular what constitutes the choice variables, and what constitutes parameters.
 - (e) Provide an argument why in the present utility maximization problem, we can restrict attention to the case where the budget constraint holds as an equality, that is, we can restrict attention to the equality constraint $p_1 x_1 + p_2 x_2 = M$ (The argument should not involve the explicit computation of the optimal choices.)
2. For the utility maximization problem given above in (1) and (2) you will compute now the optimal demands.
- (a) For the first part of the problem, we will focus on *interior solutions*. Use the Lagrangian method to derive the first order conditions for the utility maximization problem. Briefly describe the optimality conditions and what do they mean.
 - (b) Graphically describe the optimality condition and carefully label the graph and all its elements.
 - (c) Explicitly compute the demand functions that come from an *interior solution*.
 - (d) Given your solution in the previous item, for what values of M , p_1 and p_2 do you get a nonsensical solution in the sense that $x_2^* < 0$? *Hint: Think about corner solutions.*
 - (e) Suppose that $p_1 = 2$, $p_2 = 3$ and $M = 6$. What is the consumer's optimal consumption bundle? Suppose now that the price of good two rises to $p_2 = 7$. What is the consumer's optimal consumption bundle? Does this corroborate your results in the previous two items?

More on next page...

3. Consider the following system of demands:

$$x_1^*(p_1, p_2, M) = \frac{M}{2p_1} + \frac{6p_2}{p_1} - 5, \quad (3a)$$

$$x_2^*(p_1, p_2, M) = \frac{M}{2p_2} + \frac{5p_1}{p_2} - 6, \quad (3b)$$

where p_1 and p_2 are the prices of good 1 (x_1) and good 2 (x_2) respectively and M is the consumer's income. Both prices and income are assumed to be strictly positive.

- (a) Compute $\frac{\partial x_1^*(p_1, p_2, M)}{\partial m}$ and $\frac{\partial x_2^*(p_1, p_2, M)}{\partial m}$. Are the goods normal or inferior?
- (b) Compute $\frac{\partial x_1^*(p_1, p_2, M)}{\partial p_2}$ and $\frac{\partial x_2^*(p_1, p_2, M)}{\partial p_1}$, i.e. the cross-partial derivatives. Are the goods complements or substitutes?
- (c) Compute $\frac{\partial x_1^*(p_1, p_2, M)}{\partial p_1}$. What is its sign?
- (d) Compute the optimal consumption bundle (x_1^*, x_2^*) when income is $M = 100$ and prices are $p_1 = 4$, $p_2 = 1$.
- (e) Compute the optimal consumption bundle (x_1^*, x_2^*) after the price of good 1 increases to $p'_1 = 7$.
- (f) Suppose that we want to compensate the consumer by giving him extra income so that he can afford at the new prices ($p'_1 = 7$) his original consumption bundle (the one in which $p_1 = 4$). How much money do we need to give the consumer? After the compensation, what is the optimal consumption bundle (x_1^*, x_2^*) that the consumer will choose at the new prices and the compensated income?
- (g) Use your answers in (d)–(f) to calculate the *substitution effect*, *income effect* and *total effect*. Illustrate your calculations with a correctly labeled diagram. Do your calculations corroborate the sign of the derivative you computed in part (c)?
- (h) Using everything you learned in items (a)–(g), give an economic interpretation of the consumer's behavior after the price increase.